

1 / 20 | Ages 5-6 | 20 minutes

## Strong paper bridges

### What we are practising

Change a structure's shape and compare how it carries a load.

### What you need

Two identical paper sheets / Two equal-height books / Large lightweight toy car

Use a large empty cardboard box as a light load.

### Before you begin

Place books about a hand-width apart on the floor.

Build low on the floor; an adult handles heavy books and the bridge never supports a person.

### Try it together

1. Lay a flat sheet across the books and gently place the toy in the middle.
2. Fold the second sheet into accordion pleats and span the same gap.
3. Test the same toy in the same position and compare bending.

### Talk, draw or show your idea

Did changing the shape change the result?

### Space to explore

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2 / 20 | Ages 5-6 | 15 minutes

## Which cloth drinks water?

### What we are practising

Observe absorption while keeping starting conditions similar.

### What you need

Cotton cloth / Smooth plastic lid / Tray / Teaspoon / Small cup of water

Compare paper towel with an unbreakable plate.

### Before you begin

Put two dry surfaces on a tray; an adult measures the water.

An adult supervises water, keeps it from electrical items, wipes spills and empties the cup afterwards.

### Try it together

1. Predict which surface will soak up water.
2. Put one teaspoon of water on each surface and wait a minute.
3. Look for spreading dampness or a separate puddle, then discuss a useful cleaning material.

### Talk, draw or show your idea

Which material would you choose for a small spill?

### Space to explore

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3 / 20 | Ages 5-6 | 15 minutes

## Leaf observer notebook

### What we are practising

Describe similarities and differences based on observation.

### What you need

Paper / Pencil / Two safe plants viewed with an adult

Use clear leaf photographs in a book.

### Before you begin

Choose familiar non-irritating plants; look without picking or tasting.

An adult chooses plants and location; no tasting, picking unknown plants or touching sap. Wash hands after handling plants.

### Try it together

1. Sketch one leaf's outline and visible lines.
2. Observe a second leaf and sketch it beside the first.
3. Circle one similarity and describe two differences such as width or edge shape.

### Talk, draw or show your idea

Which details can we actually see rather than guess?

### Space to explore

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4 / 20 | Ages 5-6 | 15 minutes

# Let the light through

## What we are practising

Observe that different materials transmit different amounts of light.

## What you need

Torch / Clear unbreakable lid / Baking paper / Cardboard / Wall

Use daylight at a window and look at objects through each material without looking at the sun.

## Before you begin

An adult holds the torch aimed at the wall at a fixed distance.

An adult controls the torch; never aim at eyes or look at the sun. Use no glass or plastic bags.

## Try it together

1. Hold the clear lid in the beam and observe the wall.
2. Replace it with baking paper and then cardboard in the same position.
3. Describe which lets a clear beam through, scatters it, or blocks it.

## Talk, draw or show your idea

Which would make a better light-blocking cover?

## Space to explore

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5 / 20 | Ages 5-6 | 10 minutes

## Feel a humming sound

### What we are practising

Notice vibration associated with making a voice sound.

### What you need

No materials; a paper sheet is optional for notes

Watch an adult demonstrate and describe what they feel instead of touching your own throat.

### Before you begin

Sit comfortably; participation and touching are optional.

Touch only your own throat lightly; never press, hold breath or force sound. Stop with any discomfort.

### Try it together

1. Rest fingertips lightly on your own throat without pressing.
2. Hum gently for a short moment and notice any vibration.
3. Stop and breathe normally; compare and repeat once if comfortable.

### Talk, draw or show your idea

What changed when you stopped humming?

### Space to explore

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6 / 20 | Ages 5-6 | 10 minutes

## The face in the mirror

### What we are practising

Recognise an image made by reflected light.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Draw a person facing a mirror in a lit room.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Draw a person facing a mirror in a lit room.
2. Ask: Is the face in the mirror another person behind the glass, or a reflected image?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Draw a person facing a mirror in a lit room. Is the face in the mirror another person behind the glass, or a reflected image?

### Space to explore

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7 / 20 | Ages 5-6 | 10 minutes

## A magnet's limited guests

### What we are practising

Relate magnetic attraction to material.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Draw an ordinary iron nail, a wooden spoon and a magnet; use pictures only.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Draw an ordinary iron nail, a wooden spoon and a magnet; use pictures only.
2. Ask: Which object does the magnet normally attract: the iron nail or wooden spoon?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Draw an ordinary iron nail, a wooden spoon and a magnet; use pictures only. Which object does the magnet normally attract: the iron nail or wooden spoon?

### Space to explore

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8 / 20 | Ages 5-6 | 10 minutes

## Water enters the plant

### What we are practising

Connect roots with water uptake.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Draw a plant with roots in moist soil and leaves above.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Draw a plant with roots in moist soil and leaves above.
2. Ask: Which part takes up most water from the soil in this ordinary land plant?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Draw a plant with roots in moist soil and leaves above. Which part takes up most water from the soil in this ordinary land plant?

### Space to explore

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9 / 20 | Ages 5-6 | 10 minutes

## The sprout photo order

### What we are practising

Order observed plant development stages.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Draw a bean seed, the same seed with a tiny root, and a seedling with leaves.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Draw a bean seed, the same seed with a tiny root, and a seedling with leaves.
2. Ask: Put these stages in growing order. Which comes between seed and leafy seedling?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Draw a bean seed, the same seed with a tiny root, and a seedling with leaves. Put these stages in growing order. Which comes between seed and leafy seedling?

### Space to explore

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10 / 20 | Ages 5-6 | 10 minutes

## From caterpillar to butterfly

### What we are practising

Recognise stages in an insect life cycle.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Draw butterfly stages: egg, caterpillar, pupa, adult butterfly.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Draw butterfly stages: egg, caterpillar, pupa, adult butterfly.
2. Ask: Which stage lies between caterpillar and adult butterfly?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Draw butterfly stages: egg, caterpillar, pupa, adult butterfly. Which stage lies between caterpillar and adult butterfly?

### Space to explore

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11 / 20 | Ages 5-6 | 10 minutes

## Six legs or eight?

### What we are practising

Use a body feature to compare animal groups.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Draw an adult ant with six legs and a spider with eight; make every leg visible.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Draw an adult ant with six legs and a spider with eight; make every leg visible.
2. Ask: Which pictured animal is an insect?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Draw an adult ant with six legs and a spider with eight; make every leg visible. Which pictured animal is an insect?

### Space to explore

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12 / 20 | Ages 5-6 | 10 minutes

## Water borrows a shape

### What we are practising

Distinguish liquid shape from amount.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Draw the same amount of water in a wide bowl and in a narrow cup.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Draw the same amount of water in a wide bowl and in a narrow cup.
2. Ask: Does liquid water keep the bowl's wide shape after being poured into the cup?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Draw the same amount of water in a wide bowl and in a narrow cup. Does liquid water keep the bowl's wide shape after being poured into the cup?

### Space to explore

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13 / 20 | Ages 5-6 | 10 minutes

## The downhill rain trail

### What we are practising

Predict flow using a slope.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Draw a sloping path with a puddle at its lower end and arrows for flowing rainwater.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Draw a sloping path with a puddle at its lower end and arrows for flowing rainwater.
2. Ask: Without a pump or other push, which way does the water generally run along this slope?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Draw a sloping path with a puddle at its lower end and arrows for flowing rainwater. Without a pump or other push, which way does the water generally run along this slope?

### Space to explore

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14 / 20 | Ages 5-6 | 10 minutes

## Nearer to the quiet bell

### What we are practising

Relate distance to perceived loudness.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Draw one bell with two listeners in an open quiet place, one nearby and one farther away.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Draw one bell with two listeners in an open quiet place, one nearby and one farther away.
2. Ask: With the same bell ringing, who would generally hear a louder sound?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Draw one bell with two listeners in an open quiet place, one nearby and one farther away. With the same bell ringing, who would generally hear a louder sound?

### Space to explore

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15 / 20 | Ages 5-6 | 10 minutes

## Leaves and light

### What we are practising

Distinguish plant food-making from animal feeding.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Draw a green leafy plant near suitable daylight. This is a drawing activity.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Draw a green leafy plant near suitable daylight. This is a drawing activity.
2. Ask: Do most green plants get all their food by eating soil, or make sugars using light?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Draw a green leafy plant near suitable daylight. This is a drawing activity. Do most green plants get all their food by eating soil, or make sugars using light?

### Space to explore

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16 / 20 | Ages 5-6 | 10 minutes

## A Moon in the daytime

### What we are practising

Replace a common sky assumption with a correct possibility.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Draw a pale Moon in a daytime sky; this is a picture, not a Sun-viewing activity.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Draw a pale Moon in a daytime sky; this is a picture, not a Sun-viewing activity.
2. Ask: Is seeing the Moon during the day possible?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Draw a pale Moon in a daytime sky; this is a picture, not a Sun-viewing activity. Is seeing the Moon during the day possible?

### Space to explore

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17 / 20 | Ages 5-6 | 10 minutes

## Sugar you cannot see

### What we are practising

Distinguish dissolving from disappearing entirely.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Imagine a small spoonful of sugar stirred until fully dissolved in water; use a drawing only.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Imagine a small spoonful of sugar stirred until fully dissolved in water; use a drawing only.
2. Ask: Has the sugar ceased to exist because we can no longer see separate grains?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Imagine a small spoonful of sugar stirred until fully dissolved in water; use a drawing only. Has the sugar ceased to exist because we can no longer see separate grains?

### Space to explore

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18 / 20 | Ages 5-6 | 10 minutes

## The returning clap

### What we are practising

Explain a repeated sound using reflection.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Draw a person clapping facing a distant hard wall and hearing a delayed repeat.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Draw a person clapping facing a distant hard wall and hearing a delayed repeat.
2. Ask: Could the repeated sound be an echo of the first clap?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Draw a person clapping facing a distant hard wall and hearing a delayed repeat. Could the repeated sound be an echo of the first clap?

### Space to explore

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19 / 20 | Ages 5-6 | 10 minutes

# A puddle dries without a towel

## What we are practising

Identify evaporation in an everyday scene.

## What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

## Before you begin

Draw a small puddle that gradually shrinks on a dry day, with no drain or wiping.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

## Try it together

1. Draw a small puddle that gradually shrinks on a dry day, with no drain or wiping.
2. Ask: Where can some of its water go?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

## Talk, draw or show your idea

Draw a small puddle that gradually shrinks on a dry day, with no drain or wiping. Where can some of its water go?

## Space to explore

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20 / 20 | Ages 5-6 | 10 minutes

## Today's weather is one record

### What we are practising

Limit a conclusion to the observations available.

### What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

### Before you begin

Draw today's weather card showing rain; no other dates are recorded.

An adult reads and draws with the child using age-appropriate drawing tools. Explore the situation in pictures.

### Try it together

1. Draw today's weather card showing rain; no other dates are recorded.
2. Ask: Does one rainy-day record prove that it rains here every day?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

### Talk, draw or show your idea

Draw today's weather card showing rain; no other dates are recorded. Does one rainy-day record prove that it rains here every day?

### Space to explore

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## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 1. Strong paper bridges

#### A little hint

Keep the gap and load the same for both tests.

#### Answer / what to notice

A pleated sheet often bends less; record your actual result if both hold or both collapse.

Folds can make paper resist bending without adding more paper.

#### Make it easier

An adult folds while the child tests and observes.

#### Take it further

Try another fold shape with a fresh identical sheet.

#### Another day, away from the screen

Look for folded or ridged cardboard in packaging.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 2. Which cloth drinks water?

#### A little hint

Check whether the water stays separate or enters the material.

#### Answer / what to notice

Cotton usually absorbs water; smooth plastic keeps it on the surface. Coatings can change results.

Absorbent materials take water into small spaces within them.

#### Make it easier

Observe one drop-sized splash on each surface with adult help.

#### Take it further

Repeat with dry samples and two equal spoonfuls each.

#### Another day, away from the screen

Choose a suitable cloth to help clean the tray.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 3. Leaf observer notebook

#### A little hint

Look at the outline, colour and lines on each leaf.

#### Answer / what to notice

Responses depend on the leaves; accept observations tied to visible features, not guessed plant names.

A scientific description separates what we observe from what we assume.

#### Make it easier

Compare only broad and narrow outlines.

#### Take it further

Draw a third leaf and group it using a stated feature.

#### Another day, away from the screen

Revisit one plant later and record any visible change.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 4. Let the light through

#### A little hint

Compare the brightness on the wall rather than staring at the torch.

#### Answer / what to notice

Clear plastic transmits much light; baking paper scatters some; cardboard blocks most direct light.

The material affects how much light passes through and how it spreads.

#### Make it easier

Compare only clear plastic and cardboard.

#### Take it further

Predict what two layers of baking paper will do, then test.

#### Another day, away from the screen

Find a window, curtain and wall and discuss their different jobs.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 5. Feel a humming sound

#### A little hint

Focus on a gentle buzzing feeling, not loudness.

#### Answer / what to notice

You may feel vibration during humming and less during quiet breathing; it can be subtle.

Vocal folds vibrate to produce voiced sounds such as humming.

#### Make it easier

Listen while the adult describes the sensation.

#### Take it further

Compare a gentle hum with a softly spoken vowel.

#### Another day, away from the screen

Listen for voices and instruments in a familiar song.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 6. The face in the mirror

#### A little hint

The image follows the person's movements.

#### Answer / what to notice

A reflected image.

Light from the person reflects from the mirror to our eyes.

#### Make it easier

Compare a still drawing with the moving reflection idea.

#### Take it further

Predict what happens to the image when the person steps away from the mirror.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 7. A magnet's limited guests

#### A little hint

Iron has a magnetic property that wood lacks.

#### Answer / what to notice

The iron nail.

A magnet attracts iron; it does not attract every material or every metal.

#### Make it easier

Name each object's material first.

#### Take it further

Explain why “all shiny things stick to magnets” is not a safe rule.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 8. Water enters the plant

#### A little hint

Find the part touching the soil.

#### Answer / what to notice

The roots.

Roots take up water that can move through the plant to other parts.

#### Make it easier

Trace from the soil to the roots.

#### Take it further

Draw arrows for water moving from roots up through the stem.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 9. The sprout photo order

#### A little hint

Look for a new root before the larger plant.

#### Answer / what to notice

The seed with a tiny root.

In bean germination the young root appears before the seedling develops visible leaves.

#### Make it easier

Place the seed first and seedling last.

#### Take it further

Explain why photographs need dates to tell how long growth took.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 10. From caterpillar to butterfly

#### A little hint

It is the changing stage inside a protective outer covering.

#### Answer / what to notice

The pupa.

Butterflies undergo egg, larva, pupa and adult stages; a caterpillar is the larva.

#### Make it easier

Point to each stage while naming it.

#### Take it further

Draw arrows back from adult to eggs to show the next generation.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 11. Six legs or eight?

#### A little hint

Adult insects have six legs.

#### Answer / what to notice

The ant; the spider is an arachnid, not an insect.

Counting the legs supports the distinction in these two examples.

#### Make it easier

Mark each leg once while counting.

#### Take it further

Explain why not seeing all legs in a photograph could make counting unreliable.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 12. Water borrows a shape

#### A little hint

Compare the containers' shapes.

#### Answer / what to notice

No; it takes the shape of the part of the cup it occupies.

A liquid can flow and change shape while its amount stays the same if none spills.

#### Make it easier

Trace the water's boundary in each drawing.

#### Take it further

Contrast a wooden block placed in the cup: it keeps its own shape.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 13. The downhill rain trail

#### A little hint

Find the lower end.

#### Answer / what to notice

Down the slope towards the low end.

Gravity helps water flow downhill; obstacles can redirect its route.

#### Make it easier

Mark high and low ends.

#### Take it further

Add a barrier and sketch a possible route around it.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 14. Nearer to the quiet bell

#### A little hint

The sound spreads out as it travels.

#### Answer / what to notice

The nearby listener, all else being equal.

Distance usually reduces the sound reaching a listener; walls and other sounds can affect real results.

#### Make it easier

Compare two clearly different distances.

#### Take it further

Explain why a wall between bell and listener would add another factor.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 15. Leaves and light

#### A little hint

Soil supplies water and minerals, but light has another role.

#### Answer / what to notice

They make sugars using light, water and carbon dioxide.

Green leaves commonly perform photosynthesis; soil is not food swallowed like an animal's meal.

#### Make it easier

Say plants use light to help make food.

#### Take it further

Include arrows for water and air as well as light in a diagram.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 16. A Moon in the daytime

#### A little hint

The Moon does not belong only to night-time hours.

#### Answer / what to notice

Yes; the Moon is sometimes visible during daylight.

Its position and brightness can make it visible against the daytime sky.

#### Make it easier

Compare drawings of the Moon by day and by night.

#### Take it further

Explain why “sometimes visible” does not mean visible every day at every time.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 17. Sugar you cannot see

#### A little hint

Dissolved is different from destroyed.

#### Answer / what to notice

No; it remains in the water as dissolved sugar.

The sugar spreads through the water in particles too small to see individually.

#### Make it easier

Draw tiny imaginary dots mixed through the water.

#### Take it further

Explain why clear-looking water may still contain dissolved materials.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 18. The returning clap

#### A little hint

Sound can bounce from a surface.

#### Answer / what to notice

Yes, if reflected sound returns after a noticeable delay.

An echo is reflected sound; a second person need not be clapping.

#### Make it easier

Trace a path to the wall and back.

#### Take it further

Explain why a very close wall may not give a separately heard echo.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### 19. A puddle dries without a towel

#### A little hint

Water can change into an invisible gas.

#### Answer / what to notice

Into the air as water vapour by evaporation.

Evaporation can happen without boiling; the rate depends on conditions.

#### Make it easier

Draw an arrow labelled water vapour.

#### Take it further

Compare two possible conditions that affect drying, such as moving air and still air.

#### Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

## For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

### **20. Today's weather is one record**

#### **A little hint**

One date is not every date.

#### **Answer / what to notice**

No; it tells us about the recorded day only.

More records are needed to describe a weather pattern.

#### **Make it easier**

Say “today was rainy” instead of “it always rains”.

#### **Take it further**

Design a week's weather chart with a place for each day.

#### **Another day, away from the screen**

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.